

Problem Set: Pandas for Financial Time Series

Note: All datasets necessary for these exercises can be downloaded from lecture 4.

1: i) Use Pandas to calculate the daily net returns for the S&P 500 closing levels, and compare them to the continuously compounded (logarithmic) returns by plotting each time series.

ii) Hone in on isolated periods with a homogeneous appearance. For instance, plot the returns for the relatively calm period in the mid 1990s separately, and do so for the volatile period around the turn of the century that characterizes the dot com boom and bust. One can also do this for the calm period in the mid 2000s between the dot com bust and the global financial crisis of 2008, and compare it with the volatile years of the crisis itself. For all of these periods, display both the net returns and the continuously compounded returns.

2: Carry out similar investigations as in the previous example for the Russell 2000 index. Extract the closing prices, calculate and display the net returns and the continuously compounded returns for the entire series, and isolate specific periods, as for the S&P 500 index and display them separately.

3: Carry out a similar analysis as in the previous problems for the Euro Stoxx 50 index.